

Mengmeng Wang

Data Scientist | Applied ML & GenAI — LLM Pipelines, Statistical Analysis

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Summary

Data scientist with a research background (PhD, statistical signal processing), building predictive machine learning models, conducting statistical analyses, and developing LLM and GenAI pipelines and systems for extracting and synthesising evidence from clinical literature and unstructured research data. Core toolkit: Python, R, SQL — machine learning, explainable AI, and applied GenAI, including local LLM deployment for secure, scalable text analysis.

Skills

Machine Learning & Explainable AI Predictive modelling (XGBoost), explainable AI (SHAP), supervised & unsupervised learning, time series analysis, feature engineering

Applied GenAI & LLM local LLM deployment (Ollama), RAG, Hugging Face Transformers, embeddings, topic modelling (BERTopic), structured extraction pipelines

Programming & Data Python (pandas, numpy, scikit-learn, FastAPI), R, SQL, MATLAB, Azure, Git version control

Experience

Data Scientist, *Orygen, Centre for Youth Mental Health*

Feb. 2023 – Present

- Build predictive machine learning models and conduct statistical analyses on large, complex clinical and survey datasets — including a multimodal pipeline processing 175,000+ records (XGBoost, 0.86 ROC-AUC).
- Design and deliver applied GenAI systems, including LLM pipelines for evidence extraction and retrieval-based synthesis from clinical literature and unstructured research data — using AI-assisted development to accelerate delivery while owning architecture and validation.
- Work across the full pipeline — data cleaning, feature engineering, modelling — and communicate results via dashboards, reports, and academic publications to technical and non-technical stakeholders.

Data Science & Machine Learning Tutor (Casual), *The University of Melbourne*

Mar. 2019 – Nov. 2022

- Delivered 100+ practical tutorials across undergraduate and postgraduate courses in machine learning, data processing, and signals and systems — covering data wrangling, NLP, supervised and unsupervised learning, and neural networks, with practical Python implementation.

Consulting Engineer, *China Academy of Information and Communications Technology*

May 2015 – Apr. 2017

- Analysed domestic and international trends in telecommunications and information technology, producing 20+ consulting reports and data analysis deliverables for government and industry clients.

Selected Projects

Applied GenAI: Evidence Extraction & Cross-Paper Synthesis

- Designed an LLM-based pipeline extracting structured data from clinical trial literature using constrained, schema-driven prompts on a locally hosted model, achieving 92% successful extraction across a 600+ paper pilot corpus.
- Extended the system with a cross-paper synthesis layer — vector retrieval, reranking, and map-reduce narrative synthesis — to answer evidence questions spanning multiple papers rather than one document at a time.

Retrieval-Augmented Generation System for Internal Training Resources

- Architected an end-to-end RAG application: web crawler and document ingestion, heading-aware chunking, hybrid dense + BM25 retrieval with reciprocal rank fusion, and cross-encoder reranking.
- Delivered grounded, citation-backed answers via a FastAPI backend with streaming responses; supports both local LLM inference and cloud API fallback.

Multimodal Outcome Prediction from Linked Multi-Source Data

- Built a multimodal XGBoost pipeline integrating structured clinical data with text-derived features from unstructured free-text, predicting outcomes across 175,000+ records with ROC-AUC of 0.86.
- Constructed geospatial and temporal features — including haversine distances from linked SA1 geodata and rolling history counts (30- and 90-day windows) — executed on an Azure cloud virtual machine.

Education

Doctor of Philosophy (PhD) — Statistical Signal Processing, *The University of Melbourne*

2017 – 2023

Master of Science (MSc), *University of Bristol*

2013 – 2014

Bachelor of Engineering (BEng), *Beijing University of Posts and Telecommunications*

2009 – 2013